BTS–III(R/S)–02.22–0019 Reg. No.

A

B.Tech. Degree III Semester Regular/Supplementary Examination Februay 2022

ME 19-205-0305 METALLURGY AND MATERIAL SCIENCE

(*2019 Scheme*)

Time: 3 Hours Maximum Marks: 60

PART A

(Answer *ALL* questions)

$(8 \times 3 = 24)$

I. (a) Obtain the atomic packing factor and coordination number for a simple cubic crystal structure.

(b) Explain the Fick's laws of diffusion.

(c) Is iron an allotropic metal? Explain.

(d) What is induction hardening?

(e) Describe Griffith's theory of brittle fracture.

(f) What is work hardening? What is the effect of work hardening on mechanical properties of a metal?

(g) Discuss the compositions and uses of any one of copper alloys.

(h) What are some of the properties of alloy steels that make them important?

PART B

$(4 \times 12 = 48)$

II. (a) Calculate the unit cell dimension and radius of iridium atom, having FCC crystal structure, a density of 22.4 gm/cc and atomic weight of 192.2 gm/mol. (6)

(b) Differentiate between edge dislocation and screw dislocation with neat sketches. (6)

OR

III. (a) What are Miller Indices? Sketch (101), (110), (011), $\left[\bar{1}20\right]$, $\left[2\bar{1}\bar{2}\right]$ on a cubic unit cell. (6)

(b) Briefly explain nucleation and crystal growth. (6)

IV. (a) Explain Gibb's phase rule and apply it at the eutectic point of Pb-Sn alloy system. (3)

(b) Draw the iron carbon equilibrium diagram, label it and write the reactions occurring at the invariant points indicating the temperature and composition of each phase. (9)

OR

V. (a) Justify the need of Heat treatment processes for metals. Explain with neat sketch TTT diagram for heat treatment of steel. (6)

(b) What is hardenability? Explain the Jominy end quench test. (6)

VI. (a) Explain mechanisms of plastic deformations of metal by slip and twinning. (6)

(b) Explain hot working and cold working with examples. (6)

OR

(P.T.O.)

VII. (a) Sketch creep curve and explain different stages of creep. (4)

(b) What do you mean by ductile fracture? With the help of neat sketches explain the various stages of ductile fracture. (5)

(c) List any three factors affecting the Fatigue. (3)

VIII. (a) Comment on shape memory alloys and biomaterials. (4)

(b) What are ceramics? List the various applications of ceramics. (4)

(c) List two applications of composites. Explain about any two types of composites. (4)

OR

IX. (a) Write the composition, application and properties of any three alloys of ferrous metals? (6)

(b) Discuss the applications of aluminium and magnesium alloys. (6)

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